

Sai GopalRathod

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SUMMARY

Software developer and applied AI engineer with 4+ years of experience building RAG pipelines, LLM-powered APIs, and distributed data systems across Amazon, Altice USA, and NGA Group. Delivered end-to-end RAG architectures using LangChain, OpenAI APIs, Pinecone, and FAISS, with verifiable accuracy gains (62% to 88%) and measurable business impact (35% reduction in ad-hoc analyst requests, 60% ticket automation). Strong backend foundation in Python, Flask, FastAPI, PostgreSQL, AWS, and GCP.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, Bash/Shell Scripting, Java, C++

AI & LLMs: OpenAI APIs, LangChain, LlamaIndex, Prompt Engineering (system prompts, few-shot, guardrails), RAG pipeline design (chunking strategies, embedding model selection, reranking), Vector Databases (Pinecone, FAISS), Evaluation (faithfulness, retrieval precision, answer relevance), Fine-Tuning Concepts

Backend & Frameworks: Flask, FastAPI, Django, RESTful API Design, Microservices, SQLAlchemy

Cloud & DevOps: AWS (EC2, S3, Lambda, Glue, Athena), GCP (BigQuery, Vertex AI), Docker, Kubernetes, CI/CD, Git

Databases: PostgreSQL, MySQL, MongoDB, Snowflake, Redis

ML & NLP: NLP, XGBoost, Random Forest, SHAP Analysis, Anomaly Detection, PyTorch, Scikit-learn, Pandas, NumPy, Model Evaluation (ROC-AUC, Cross-Validation)

PROFESSIONAL EXPERIENCE

AI/ML Engineer | **NGA Group Inc.** | North Brunswick Township, NJ

Jul 2024 to Present

- Built a production RAG assistant on internal reporting data using OpenAI APIs, LangChain, and a Pinecone vector store; enabled non-technical stakeholders to query analytics in natural language and reduced ad-hoc analyst requests by 35%.
- Designed and iterated on chunking strategy, embedding model selection, and prompt engineering templates (system prompts, few-shot examples, guardrails) with structured evaluation criteria (faithfulness, answer relevance, retrieval precision), lifting LLM response accuracy from 62% to 88% on a curated ground-truth test set.
- Owned data architecture for 3 production databases across 5+ concurrent projects, defining schema design, indexing strategy, and integrity constraints that reduced downstream engineering corrections by 15%.
- Designed 10+ Python and SQL automation workflows replacing entirely manual reporting processes, recovering 20 hours per month of analyst capacity and improving data freshness to near real time.
- Applied statistical profiling across production logs to isolate persistent bottlenecks, implemented targeted fixes achieving a 10% performance gain and 25% reduction in critical errors with full regression coverage.

AI/ML Engineering Intern | **Altice USA** | New York, NY

Jan 2024 to May 2024

- Built a Python-based anomaly detection service from scratch using Isolation Forest with threshold-based fallbacks, improving hardware fault detection accuracy by 25% and increasing network reliability by 20% across a subscriber base of millions.
- Developed and evaluated ML model pipelines on GCP (BigQuery, Vertex AI), contributing to team-wide migration from legacy analytics tooling through structured enablement sessions and hands-on workshops across 3+ teams.
- Redesigned 8 ETL pipelines in Palantir Foundry with fully automated, idempotent workflows, reducing processing time by 30% and establishing data reliability standards adopted across the product organization.
- Designed and maintained A/B testing infrastructure for ML-driven product feature evaluations, introducing statistical rigor to rollout decisions previously based on intuition.
- Collaborated with data and product teams to surface model insights, translating outputs into actionable recommendations and supporting cross-functional alignment on AI adoption priorities.

Programmer Analyst | **Amazon** | Hyderabad, India

Jul 2020 to Dec 2022

- Architected an end-to-end data quality system using AWS Glue and Python across 12 interdependent datasets: automated profiling, schema validation, and anomaly flagging that improved pipeline efficiency by 41% and reduced system downtime by 15%.
- Designed forecasting models in Python and Amazon Athena managing 500,000+ records; improved demand planning accuracy by 10% and contributed to an 8% quarterly revenue uplift by identifying demand signal patterns the incumbent model was ignoring.
- Standardized cross-functional data definitions and delivery cadence across 4 teams, cutting reporting turnaround by 25%.

KEY PROJECTS

AI-Powered Customer Support Automation Platform | Python, Flask, FastAPI, OpenAI API, LangChain, RAG, FAISS, PostgreSQL, SQLAlchemy, NLP

- Designed full system architecture independently: Flask/FastAPI service layer with RESTful endpoints, LLM-driven intent classification, a RAG layer using FAISS embeddings over the support knowledge base, a PostgreSQL and SQLAlchemy-backed SQL generation engine, and a human escalation fallback for low-confidence classifications.
- Automated approximately 60% of inbound ticket volume, reducing average wait time by 40%. Awarded 3rd Prize at the Altice USA Hackathon; referenced internally as a model for AI-driven support automation.

Predictive Campaign Intelligence System | Python, XGBoost, Random Forest, SHAP, Tableau

- Designed EDA pipeline across 45,000 customer records, feature engineering from transactional signals, model selection, and a SHAP interpretability layer for non-technical stakeholders. Final architecture: XGBoost and Random Forest ensemble (ROC-AUC: 0.72). Dashboard adopted by marketing leadership, improving conversion rate predictions by 18%.

EDUCATION

MS in Business Analytics | Hofstra University | GPA: 3.94/4.0

Jan 2023 to May 2024

B.Tech in Computer Science and Engineering | NIT Calicut | GPA: 3.0/4.0

Jul 2016 to Jun 2020